

Physics Mock Test

Electromagnetic Waves

Class 12 | UNIT 5

Marks: 24

Time: 60 Minutes

Overview: Displacement Current, Maxwell's Equations, Properties of EM Waves, and the Electromagnetic Spectrum.

Instructions: For calculations, use $c = 3 \times 10^8$ m/s and $\epsilon_0 = 8.85 \times 10^{-12}$ F/m.

Section 1: Multiple Choice Questions (1 Mark Each)

1. Nature of EM Waves

In vacuum, electromagnetic waves consist of oscillating electric and magnetic fields. These waves are fundamentally:

- a) Longitudinal
- b) Transverse
- c) Scalar
- d) Stationary

2. Universal Speed

The theoretical speed of electromagnetic waves in vacuum, as predicted by Maxwell, is approximately:

- a) 3×10^8 m/s
- b) 3×10^6 m/s
- c) 3×10^{10} m/s
- d) $1/\sqrt{(\mu_0\epsilon_0)}$

3. Frequency Trends

In the electromagnetic spectrum, which of the following radiation types possesses the highest frequency (and hence highest photon energy)?

- a) Radio waves
- b) Microwaves
- c) X-rays
- d) Infrared

4. Displacement Current

The concept of displacement current was introduced by James Clerk Maxwell to resolve a logical inconsistency in:

- a) The Photoelectric Effect
- b) Ampere's Circuital Law
- c) Ohm's Law
- d) Coulomb's Law

5. Maxwell's Laws

Which one of Maxwell's equations is a direct mathematical representation of Faraday's Law of Induction?

- a) $\nabla \cdot \mathbf{E} = \rho / \epsilon_0$
- b) $\nabla \cdot \mathbf{B} = 0$
- c) $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$
- d) $\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \partial \mathbf{E} / \partial t$

6. Energy Transport

The Poynting Vector ($\mathbf{S}$) describes the rate and direction of energy transport in an EM wave. Its magnitude represents:

- a) Energy density
- b) Intensity (Energy flux)
- c) Radiation pressure
- d) Charge flow

Section 2: Numerical & Short-Answer (2-3 Marks Each)

7. Wavelength Calculation

An electromagnetic wave oscillates with a frequency of 5×10^{14} Hz. Calculate its wavelength in vacuum. Which part of the spectrum does this belong to?

8. Spectral Identification

A wave has a wavelength of 10^{-3} m. Identify the region of the electromagnetic spectrum it belongs to and state one common practical application of this radiation.

9. Displacement Current in a Capacitor

A parallel-plate capacitor has plates of area 100 cm^2 and a separation of 1 mm. If the potential difference between the plates is changing at a rate of 10 V/s, find the magnitude of the displacement current.

10. Gauss's Law for Magnetism

State Gauss's Law for magnetism in integral or differential form. What is the physical significance of the fact that the total magnetic flux through any closed surface is always zero?

11. Energy Density Calculation

The RMS value of the electric field of an EM wave is 10 V/m. Calculate the total average energy density of the wave in vacuum.

12. Poynting Vector Magnitude

At a point in space, an EM wave has an electric field amplitude of 100 V/m and a magnetic field amplitude of 3.33×10^{-7} T. Calculate the magnitude of the Poynting vector.

Answer Key & Solutions

1. b) Transverse

Electric and Magnetic fields oscillate perpendicular to each other and to the direction of propagation.

2. a) $3 \times 10^8 \text{ m/s}$ (Accept 'd' as well as it is the exact definition).

3. c) X-rays

Order: Radio < Microwave < Infrared < Visible < UV < X-ray < Gamma.

4. b) Ampere's Circuital Law

Ensures continuity of current between capacitor plates.

5. c) $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$

Relates time-varying magnetic fields to induced electric fields.

6. b) Intensity (Energy flux)

The unit is Watts per square meter (W/m^2).

7. $6 \times 10^{-7} \text{ m}$ (600 nm) | Visible Light

$$\lambda = c / f = (3 \times 10^8) / (5 \times 10^{14}) = 0.6 \times 10^{-6} = 600 \text{ nm}.$$

8. Microwaves | Radar/Communication/Ovens

$10^{-3} \text{ m} = 1 \text{ mm}$. Microwaves range from 1 mm to 1 m.

9. $8.85 \times 10^{-10} \text{ A}$

$$I_d = C (dV/dt) = (\epsilon_0 A/d) \cdot (dV/dt) = (8.85 \times 10^{-12} \times 0.01 / 0.001) \times 10 = 8.85 \times 10^{-10} \text{ A}.$$

10. $\oint \mathbf{B} \cdot d\mathbf{A} = 0$

Significance: Magnetic monopoles do not exist; magnetic field lines form closed loops.

11. $8.85 \times 10^{-10} \text{ J/m}^3$

$$u_{\text{avg}} = \epsilon_0 E_{\text{rms}}^2 = (8.85 \times 10^{-12}) \times (10)^2 = 8.85 \times 10^{-10} \text{ J/m}^3.$$

12. $\approx 26.5 \text{ W/m}^2$

$$S = EB / \mu_0 = (100 \times 3.33 \times 10^{-7}) / (4\pi \times 10^{-7}) \approx 33.3 / 1.256 \approx 26.5 \text{ W/m}^2.$$

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